

Exam Date & Time: 29-May-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

**FOURTH SEMESTER B.TECH. (INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS -
MAY/JUNE 2023**

**SUBJECT: ICT 2256/IT-2256 COMPUTER ORGANIZATION & MICROPROCESSOR SYSTEMS
COMPUTER ORGANIZATION & MICROPROCESSOR SYSTEMS [ICT 2256]**

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Provide a neat diagram of 8086 architecture and explain the use of various registers of 8086 in detail. (5)
- 1B) Explain the Mode 0, Mode 1, Mode 2 operations of 8254 with necessary waveforms. (3)
- 1C) Explain the addition of four 4-bit binary numbers using carry save addition concept. (2)
- 2A) Illustrate memory and IO addressing modes available in 8086 with an example for each. (5)
- 2B) How is the block transfer method more efficient in DMA based IO data transfer? Explain. (3)
- 2C) Write an assembly language program to perform equivalent of AAM instruction. (2)
- 3A) List the advantages and disadvantages on direct mapping techniques.
A computer system uses 16-bit memory addresses. It has a 2K-byte cache organized in a direct-mapped manner with 64 bytes per cache block. Assume that the size of each memory word is 1 byte. Calculate the width of Tag, Block, and Word fields of the memory address. (5)
- 3B) Explain with the neat diagram on cascading of 8259 IC's. (3)
- 3C) Provide an assembly language program to display 2 digit hex decimal number available in the register DL. (2)
- 4A) Write an assembly language program to convert a 4-digit decimal number available in the data segment to an equivalent hexadecimal number and store it to the data segment. (5)
- 4B) Write an assembly language program using recursive procedure to solve nC_r , where n and r are both single digit numbers. (3)
- 4C) The 4-bit shifter has four data inputs, A_0 through A_3 and four data outputs, H_0 through H_3 . There are two serial inputs, one for shift left (I_L) and the other for shift right (I_R). (2)

Design according to the requirements given in Table 4C.

Table 4C

Select	output			
S	Ho	H1	H2	H3
0	I _R	A0	A1	A2
1	A1	A2	A2	I _L

- 5A) Explain and illustrate booth's algorithm for multiplying $(-7)_{10}$ and $(-5)_{10}$. Show all the steps. (5)
- 5B) Illustrate the role of following pins of 8086
i) TEST, ii) IO/M', iii) RESET (3)
- 5C) Explain the following assembler directives:
i) ENDP ii) OFFSET iii) PROC iv) DQ (2)

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